AESWARYA KRISHNAKUMAR

Introduction

Data visualisation is the process of transforming raw data into visual formats such as charts, graphs and maps, in order to make patterns, trends and relationships easy to understand. Furthermore, instead of restricting the analysis to tables or numbers, visualisation enables information to be displayed in a more interactive and meaningful way, which makes it easier to discover insights and communicate findings effectively [1].

The data set used for this assignment includes global health-related indicators derived from various sources including household surveys and health reports. It contains data for several countries for several years, with indicators such as smoking prevalence, obesity, and other behavioural and demographic health factors. Each row in the dataset is characterized by fields including Country, Year, Indicator Name, Subgroup (for example, age group), Region, Income group and the primary numeric measure Value (percentage or rate).

In this assignment, I did my pre-processing directly in Tableau. Specifically, I changed setting to Country, date to Year, and estimate to Value. I also removed unnecessary columns and this limited the dataset to only the necessary fields, simplifying the process of creating meaningful visualisations. The resulting dataset fields are cleaned and a screenshot is provided to illustrate the result of this process.

Three advanced visualisations were then created to analyse the data from different angles:
Choropleth Map: to show geographic comparisons of indicator values across countries **Line Chart**: to see changes over time with trend lines and moving averages.
Treemap: to visualize composition and show which groups/countries contribute the most to an indicator.

Tableau was the primary tool used to create these visualizations. Its advantages include simplicity of use through drag and drop functionality and the broad range of chart types, and powerful support for interactivity through filters, parameters and dashboards. However, Tableau also has some limitations, such as limited support for advanced statistical modeling, slower performance with very large datasets, and some features being limited in Tableau Public compared to Tableau Desktop.

Visualisation 1: Choropleth Map

The first visualisation is a Choropleth Map, used to display the geographic distribution of values for a health indicator by country. This type of map is helpful in determining regional differences, countries with higher or lower values, and a clear picture of world health patterns.

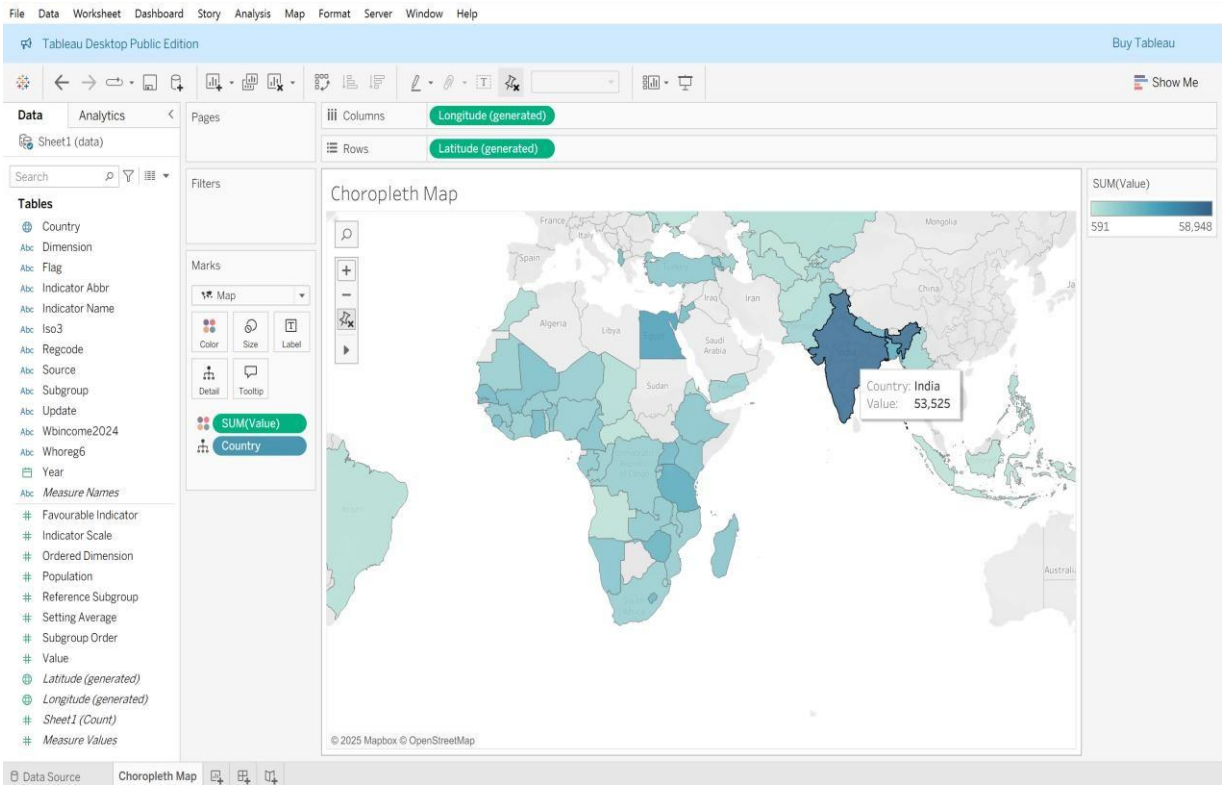

Figure 1.1 Choropleth map of health indicator values without filtering.

Figure 1.1 The first choropleth map was generated by setting Country as a geographic role of Country/Region and Value as Colour. This gave rise to a global distribution of values. However, since there was no filtering, the map contained several years and indicators at the same time, which made the map cluttered and hard to read. The result was that while patterns could be seen in some places, the view couldn't provide useful information.

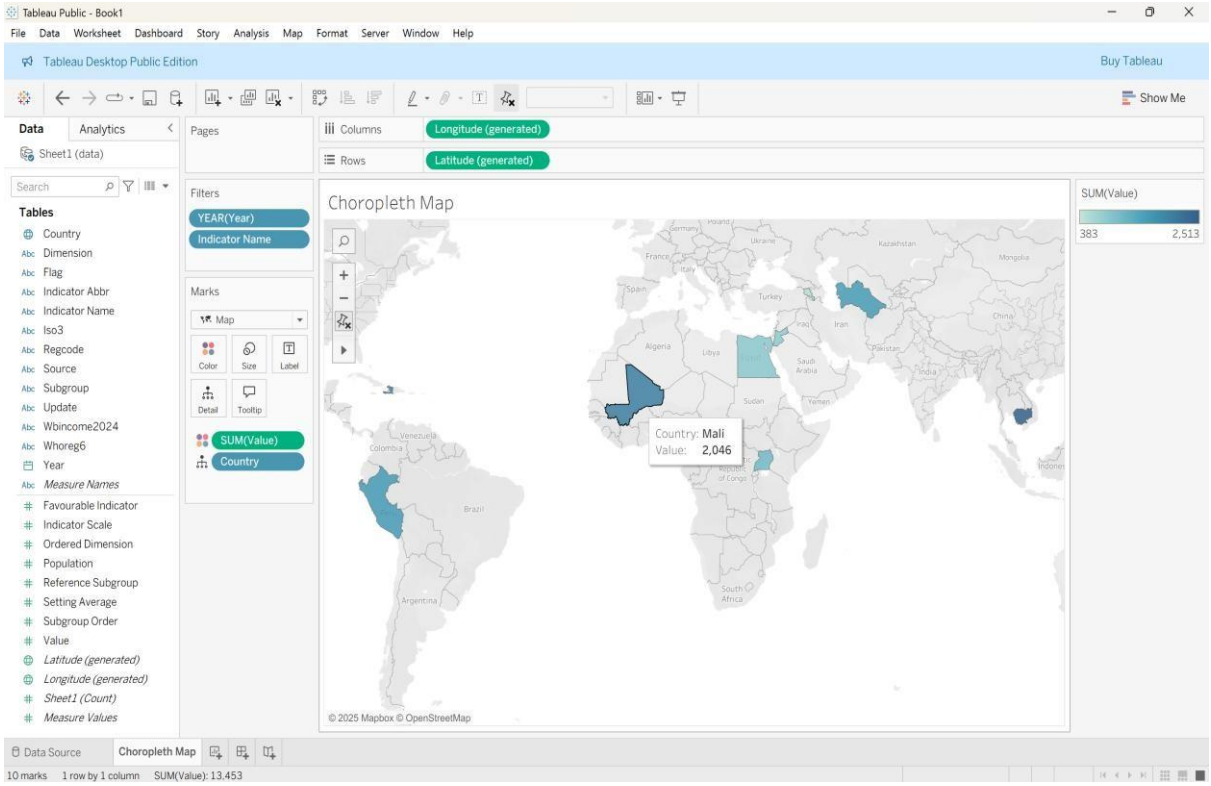

Figure 1.2 Choropleth map of health indicator values after applying Year and Indicator filters.

I used Year and Indicator Name as filters to refine the visualisation as shown in figure 1.2. This enabled me to focus on a single health metric in a single year. The result was a much clearer image of regional variation. For instance, India exhibits much higher values than many African countries in the selected year, whereas other regions such as parts of Europe exhibit relatively moderate values. This filtered map reveals important differences that were hidden in the unfiltered view, by identifying the differences in health outcomes across regions when examining a single indicator at a time.

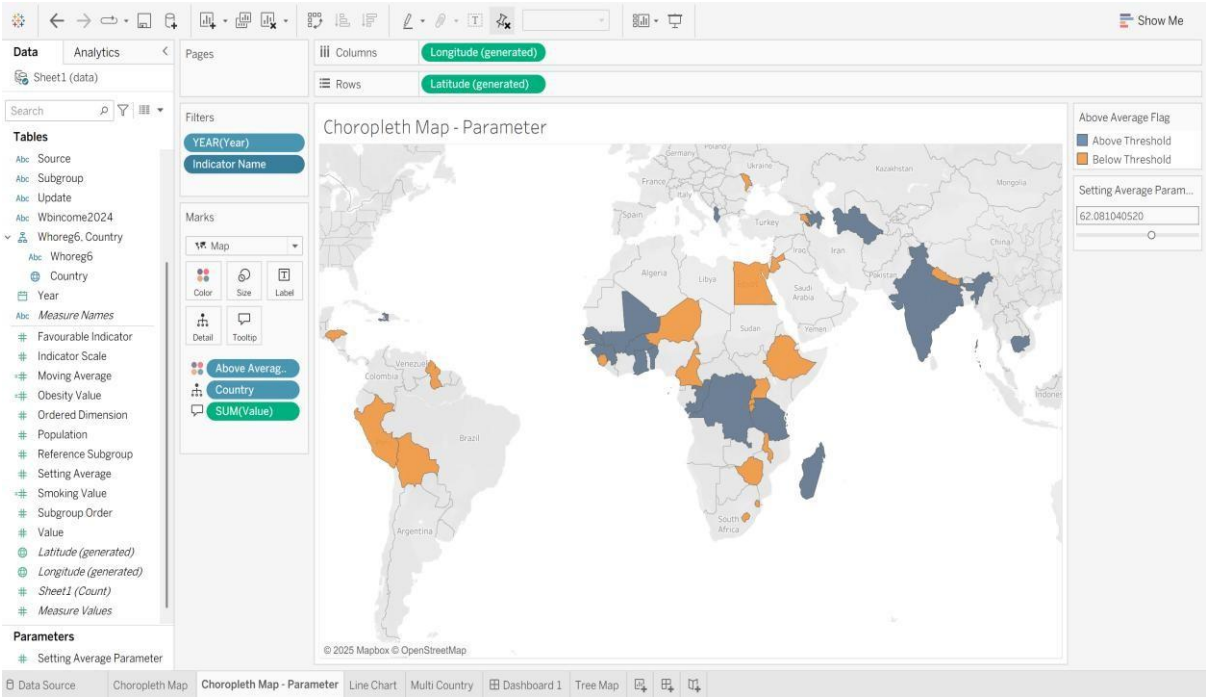

*Figure 1**Error! No text of specified style in document.**3 Choropleth Map with Parameter*

For this visualization, I took the original choropleth map and added an interactive parameter. I created a parameter called Setting Average Parameter which was Float and set to range 0 to 100, step 1. Then I created a calculated field named Above Average Flag. The choropleth map based on the parameter showed clear geographical patterns. By moving the slider, I could quickly visualize how many and which countries have a threshold value above a chosen threshold value over the 2000-2010 period. For instance, when the threshold was set at about 60, only a few countries were shown in the "Above Threshold" category, and most of the others were below. This shows regional differences, where some countries are always at the higher averages or above, while others are lagging behind.

Visualisation 2: Line Chart

A line chart is usually used to analyze time series data, such as how values change over time in years. In this assignment, it was used to analyze the development over time of selected health indicators both at the single country level and at the multi-country level. To improve interpretation, I have generated a calculated field called Moving Average in Tableau. This field was then smoothed as a statistical line to show overall trends and smooth out fluctuations. The original values are added to the moving average to give the chart both the fine detail of the yearly changes as well as the long-term direction.

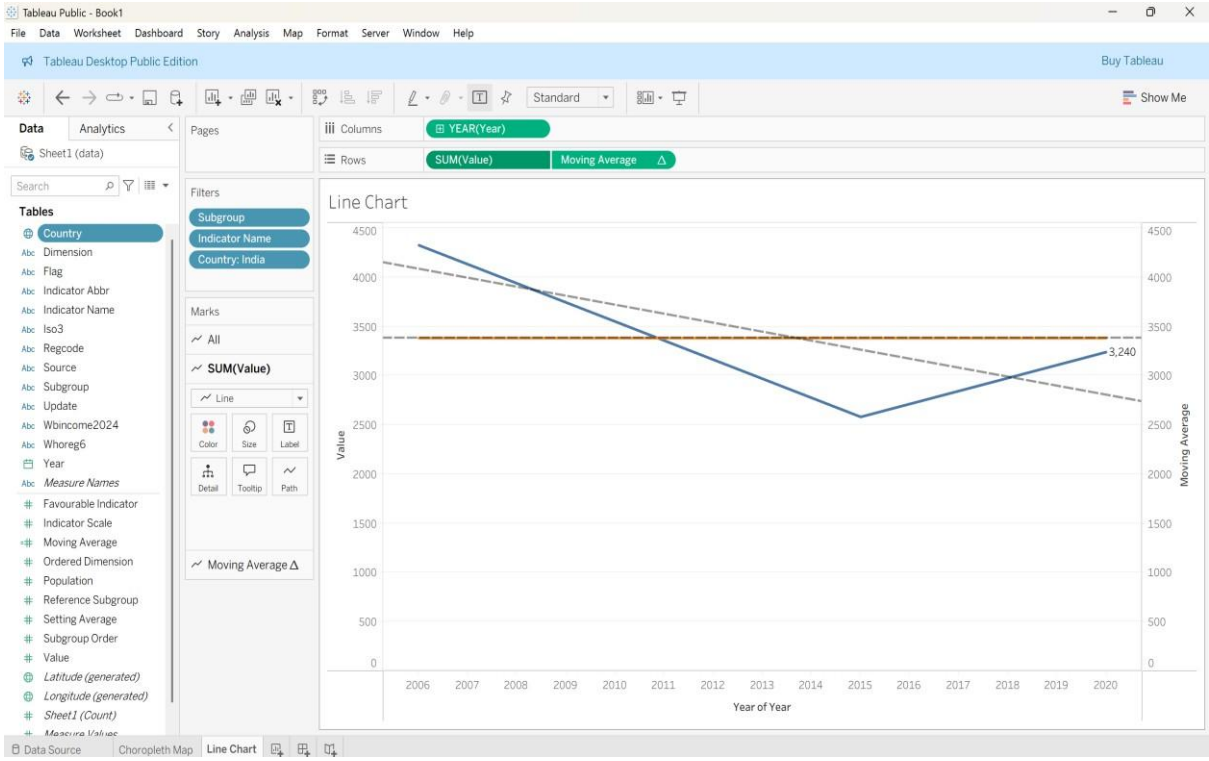

Figure 2. 1 Line chart of India with Moving Average

The line chart for India alone indicates that the indicator fell consistently until about 2015 and then started to recover in more recent years. The moving average smooths out these swings so that the overall downward-then-upward pattern is more obvious. This shows how the values of India have changed over time and yet still shows the long-term direction.

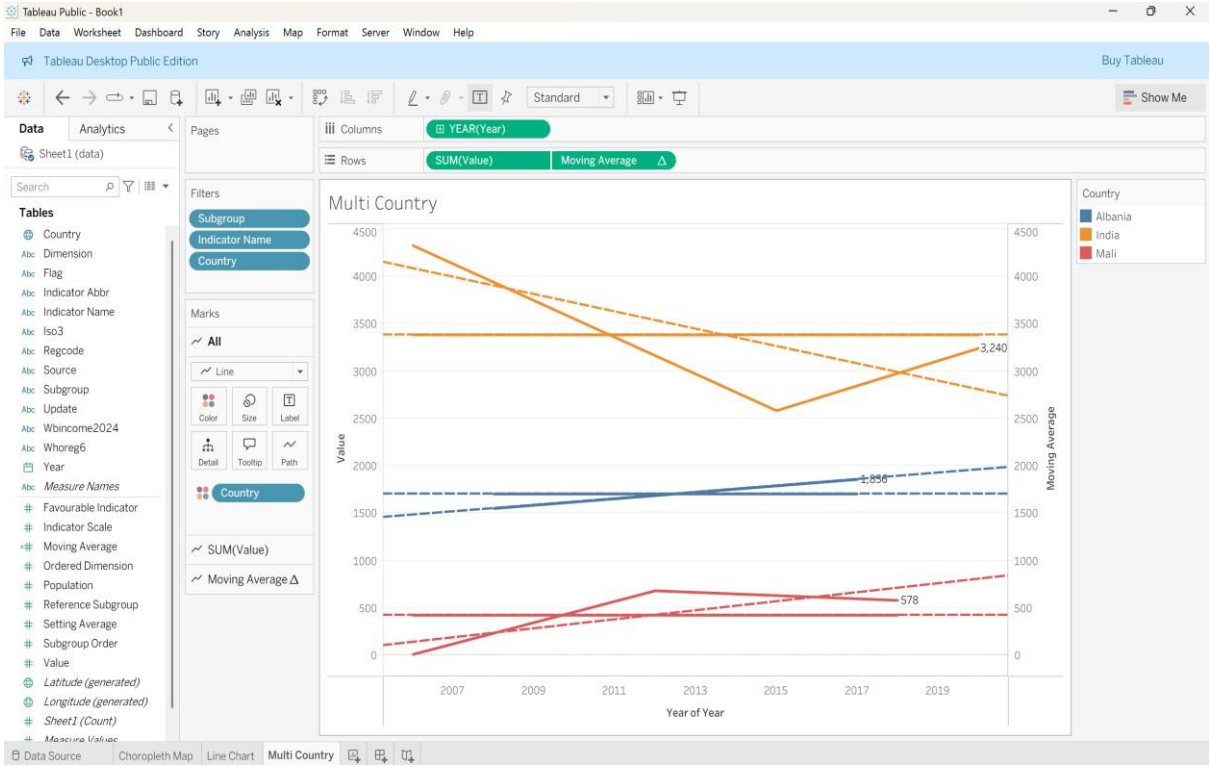

Figure 2. 2 Multi-country line chart (India, Albania, Mali)

The multi-country chart compares India, Albania and Mali. India is seen to have decreased and then recovered, Albania's values are relatively stable, and Mali shows a gradual increase over the same time period. The moving average calculated field accentuates these countrylevel differences by smoothing out noisy year-to-year fluctuations, which makes it easier to compare overall trends across countries.

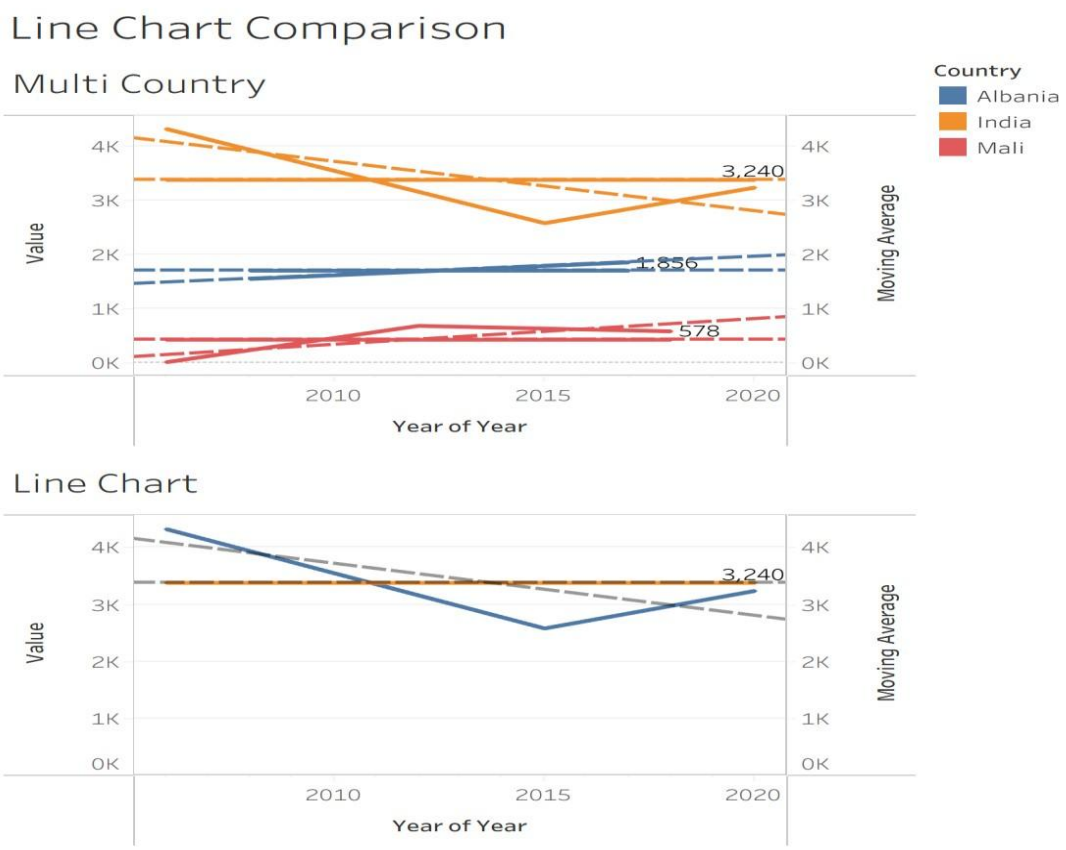

Figure 2. 3 Chart Comparison in Dashboard

As shown in figure 2c above, I created a dashboard that combines two different line charts - one that is a multi-country comparison and one that is a single country trend. The purpose of putting them together in one view is to be able to directly compare across countries, as well as to focus on individual country performance.

The first chart (Multi Country Line Chart) presents the evolution of indicator values for Albania, India and Mali between 2005 and 2020, both in terms of the actual values and in terms of the moving average. The bottom chart (Single Country Line Chart) is restricted to one country, thus providing more detail on yearly variations.

Visualization 3: Treemap of Obesity Prevalence by WHO Region and Country

For this visualization, I made a Treemap to visualize Obesity Prevalence (%) values across WHO regions and countries. The dataset was first restricted to the years 2015-2020. To further divide the analysis, I created a hierarchy between WHO Region (Whoreg6) and Country. This hierarchy meant that the treemap first showed aggregated values by region, and then drilled down into countries within the regions.

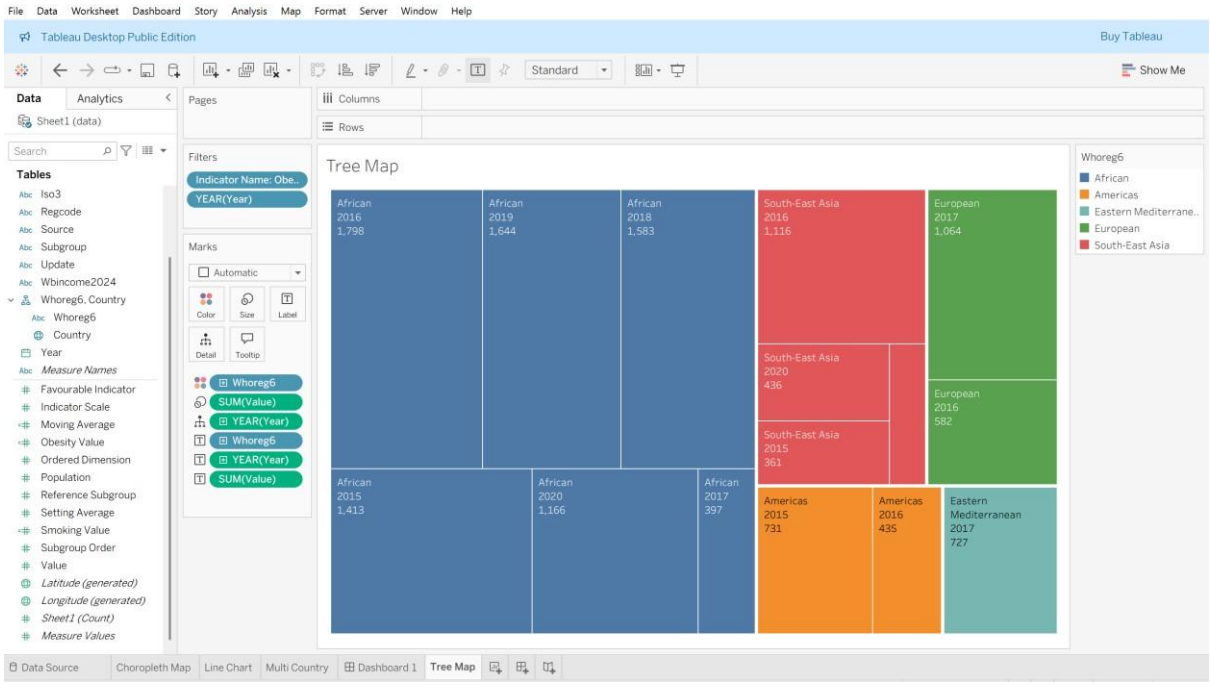

Figure 3. 1 Treemap of Obesity Prevalence by WHO Region (2015–2020)

The treemap blocks were sized by the Value field (SUM of the indicator values) and colored by WHO Region, so that the contributions of each region could be visually distinguished. Also labels to show the year and the value directly inside the blocks were added. The treemap at the regional level reveals obvious differences in obesity prevalence. Areas such as Africa take up a larger share of the treemap because of higher aggregated values, while smaller areas, such as the Eastern Mediterranean, are represented by smaller blocks.

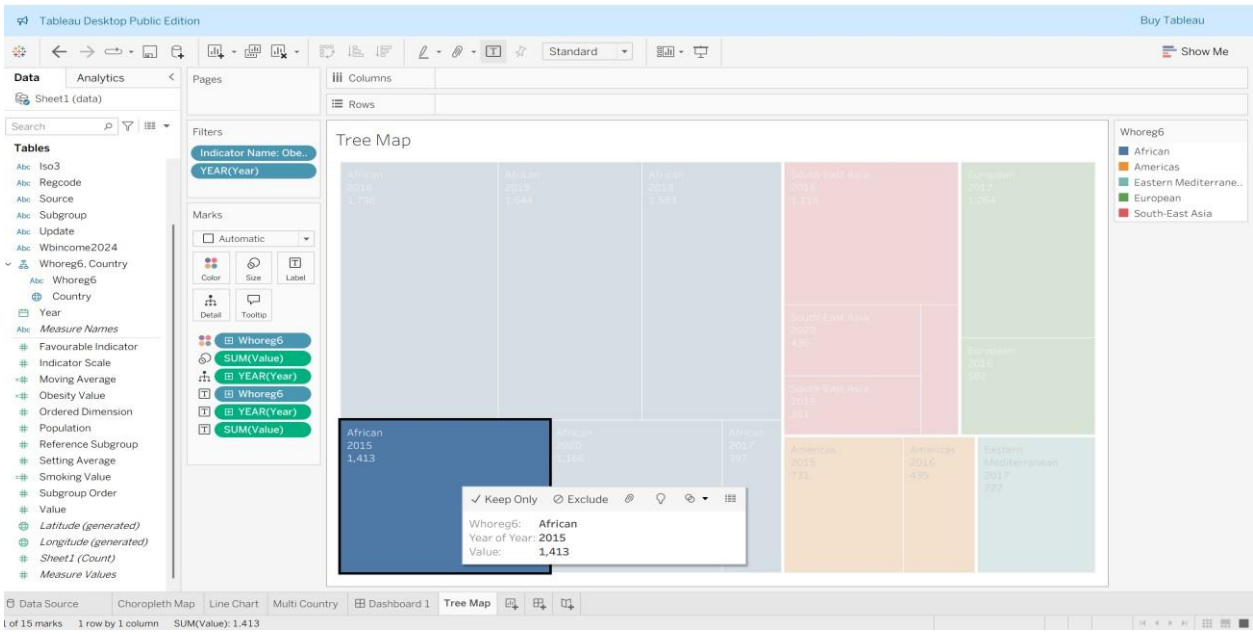

Figure 3. 2 Treemap of Obesity Prevalence by Country within the African Region (2015–2020)

Using the hierarchy, when a region is double-clicked as shown in figure above (African 2015), the treemap drills down to the country level, splitting into smaller blocks representing individual countries. This enables us to see which specific countries make the largest contribution to the regional totals. For example, in the African region, a large number of smaller countries are present, which distributes the contribution, while in other regions a few countries are responsible for the values.

The treemap thus gives both a macro-level view (comparing WHO regions) and a micro-level view (country-level contributions within regions) and is thus a powerful tool for hierarchical comparison.

Strengths and Weaknesses of the Visualisation Methods

Choropleth maps, line charts, and treemaps are the visualisation techniques used in this assignment and have strengths and weaknesses when compared to other techniques used in the literature.

Choropleth maps are very useful for illustrating geographic variation, but with respect to large countries, even if their values are low, they can look visually out of scale when compared to bar charts or tables. Line graphs are good to show changes over time and trends, but if too many countries are plotted together they can become cluttered; scatter graphs may be better to compare relationships between two variables. Treemaps are particularly appropriate for hierarchical data and proportional contributions, and are therefore better suited to show than pie charts when there are many categories. However, they can be cluttered and hard to interpret when used on very detailed datasets [2].

In terms of scalability, these methods can be used on large, multidimensional datasets, but with some caveats. Line charts with too many series are cluttered and obscure trends, and treemaps and maps lose clarity when hundreds of categories are shown at once. This means that, although these approaches work well for medium-sized multidimensional data, they will need careful filtering or aggregation when used with "big data."

Other analytical techniques are very necessary to support visualisation in the big data analysis. Data cleaning, aggregation, and dimensionality reduction (e.g., principal component analysis) can be used to simplify many of the complexity before visualisation. Automated techniques like clustering and other detection tool can identify patterns or outliers that would

be challenging to identify visually. Big data frameworks like Hadoop or Spark can be used to store and compute at scale, but can still be used in visualisation tools like Tableau with subsets or aggregations of very large datasets [3].

Conclusion

This assignment involved the use of multidimensional health indicator data which is in the form of a series of visualisations created in Tableau. The Visualisations include a choropleth map, line charts and a treemap. Each visualization gave a different view, for instance, the map showed global and regional differences, the line charts showed temporal trends within and across countries, and the treemap showed hierarchical differences between regions and their member countries.

From the analyses, a number of things became clear. Geographic maps revealed that obesity prevalence is highly uneven across WHO regions, with Africa and the Americas tending to stand out. Time series graphs illustrated the changes in obesity and smoking prevalence over the time with some countries such as India showing decreases followed by increases, and others such as Mali showing steady increases. The treemap showed both regional variation and intra-regional differences, and that aggregated regional values often conceal countrylevel differences.

Overall, Tableau was used to successfully explore large, multidimensional data. Its filtering, parameter, and drill-down capabilities allowed one to go from high-level overviews to detailed information. At the same time, there were some limitation, which includes treemaps can get cluttered with lots of small countries and line charts can smooth out short-term variation with moving averages.

References

1. Manovich, L., 2011. What is visualisation?. *Visual Studies*, *26*(1), pp.36-49.
2. Long, L.K., Hui, L.C., Fook, G.Y. and Zainon, W.M.N.W., 2017. A study on the effectiveness of tree-maps as tree visualization techniques. *Procedia Computer Science*, *124*, pp.108115.
3. Pacione, M.J., 2003. A review and evaluation of dynamic visualisation tools. *U. Strathclyde Technical Report EFoCS-50-2003*.